



Probability

Business Analytics, 2e

Learning Objectives

- LO 1** Describe concepts and the rules of probabilities.
- LO 2** Conditional probabilities and Bayes' theorem.
- LO 3** Describe the normal distribution and calculate its associated probabilities.

Probability Concepts and Probability Rules

- A **probability** is a numerical value that measures the **likelihood** that an event occurs.
 - Prob. Of an event A is a value between 0 and 1:
 $0 \leq P(A) \leq 1$.
 - 0 → impossible that event occurs
 - 1 → a definite event that always occurs
- An **experiment** is a process that leads to one of several possible outcomes.

1: Probability Concepts and Probability Rules

- There are three types of probabilities.
 1. **Subjective:** calculated by drawing on personal and subjective judgement
 - E.g. with no explicit reference, manager feels that 20% of customer feel good with the service.)
 2. **Empirical:** calculated as a relative frequency of occurrence
 - E.g. over 10 years in a city, 152 out of 180 coffee shops opened and closed within 1 year (empirical prob. = $152/180 = 0.84$)
 3. **Classical:** based on logical analysis
 - E.g. Based on logical assumption (event with equally likely), probability of rolling a “2” with a fair six-sided die as $1/6 = 0.1667$.

1. Probability Concepts and Probability Rules

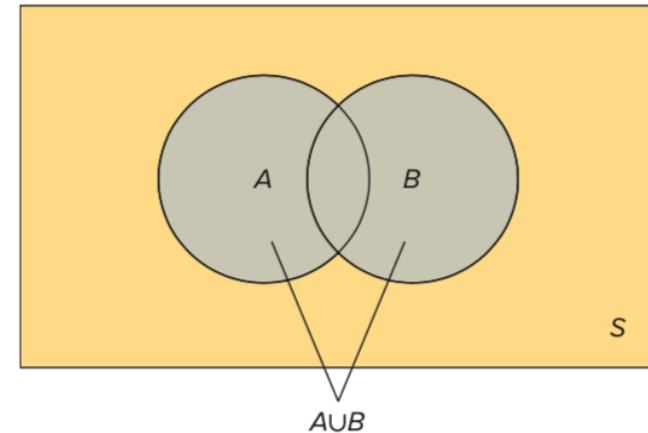
- **Sample space** of an experiment
 - Denoted S
 - Contains all possible outcomes of the experiment
 - Examples:
 - Letter grades in a course: $S = \{A, B, C, D, F\}$
 - a six-sided die: $S = \{1, 2, 3, 4, 5, 6\}$
- **An event** is any subset of outcomes of the experiment.
 - Simple event if it contains a single outcome
 - May contain several outcomes
 - Example:
 - a passing grade, $pass = \{A, B, C, D\}$
 - Roll a die, $even\ number = \{2, 4, 6\}$

Probability Concepts and Probability Rules

- Exhaustive events
 - All possible outcomes of an experiment belong to the events
 - Include all outcomes in the sample space
- Mutually exclusive events
 - They do not share any common outcomes
 - The occurrence of one event precludes the occurrence of others
- Examples:
 - Consider rolling a six-sided die. The sample space (all possible outcomes) is:
 $S=\{1,2,3,4,5,6\}$
 - Now. Let's define 2 events:
Event A: Getting an even number $\rightarrow A=\{2,4,6\}$
Event B: Getting an odd number $\rightarrow B=\{1,3,5\}$
 - Since events A and B together cover all possible outcomes of rolling the die, they are exhaustive events and since they do not share any common outcomes, they are mutually exclusive events.

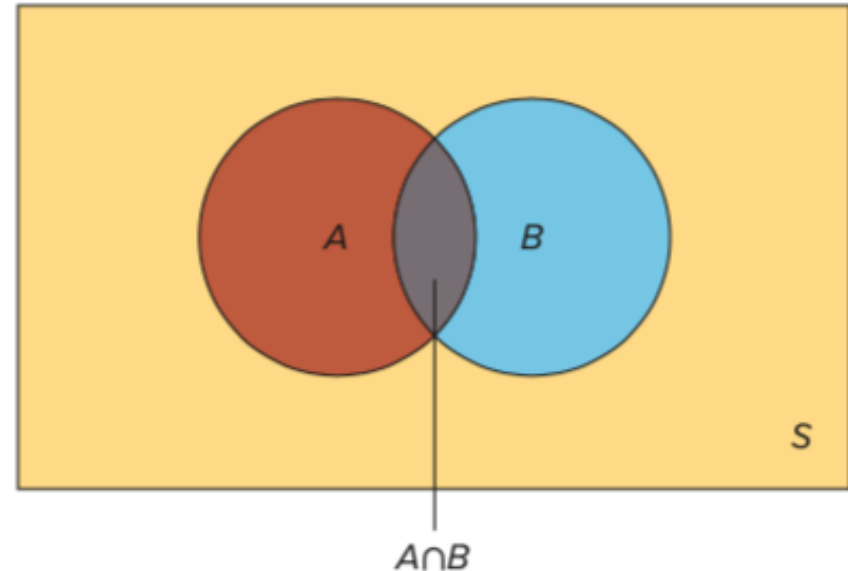
1: Probability Concepts and Probability Rules

- We can combine events to form new events.
- Venn Diagram
 - Sample space S with a rectangle
 - Two circles to represent the events A and B
- Union of two events
 - Denoted $A \cup B$
 - All outcomes in A or B (or both)



1: Probability Concepts and Probability Rules

- Intersection of two events
 - Denoted $A \cap B$
 - All outcomes in A and B
 - The portion in the Venn diagram that is included in both A and B, **the overlap**



1: Probability Concepts and Probability Rules

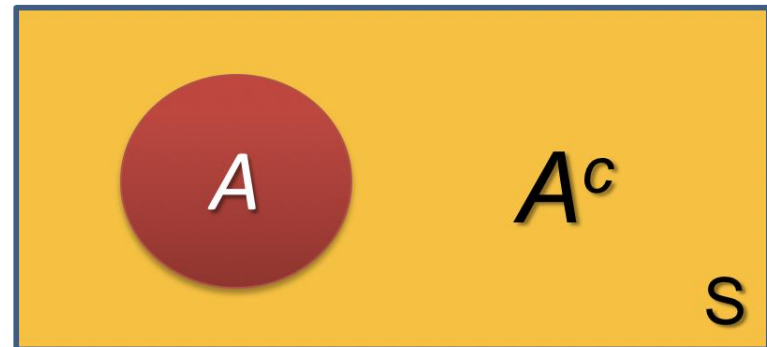
- Complement of an event A
 - Denoted A^c
 - All outcomes in the sample space S that are not in A



1: Probability Concepts and Probability Rules

- Complement rule

- The probability of the complement of an event is equal to one minus the probability of the event
- $P(A^c) = 1 - P(A)$
- This follows from $P(S) = 1$



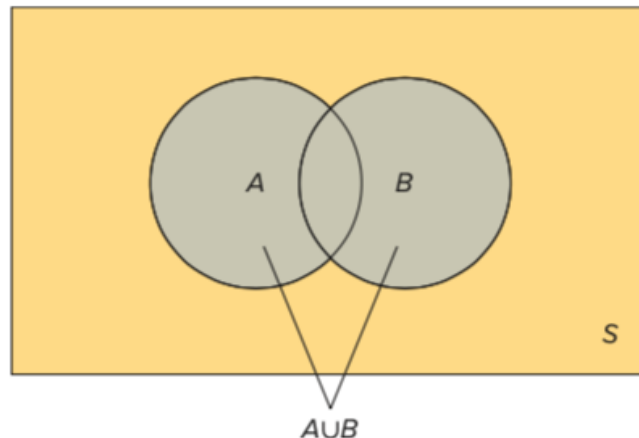
1: Probability Concepts and Probability Rules

- **Example:** 37% of female open house attendees will purchase a membership. What is the probability that a randomly selected female will not purchase a membership?
- Define A as the event that a randomly selected female will purchase a membership.
- $P(A) = 0.37$
- $P(A^c) = 1 - P(A) = 1 - 0.37 = 0.63$

1: Probability Concepts and Probability Rules

- **Addition rule**

- Used to find the probability of the union of two events
- The probability that A or B occurs, or that at least one of these events occurs
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$



1: Probability Concepts and Probability Rules

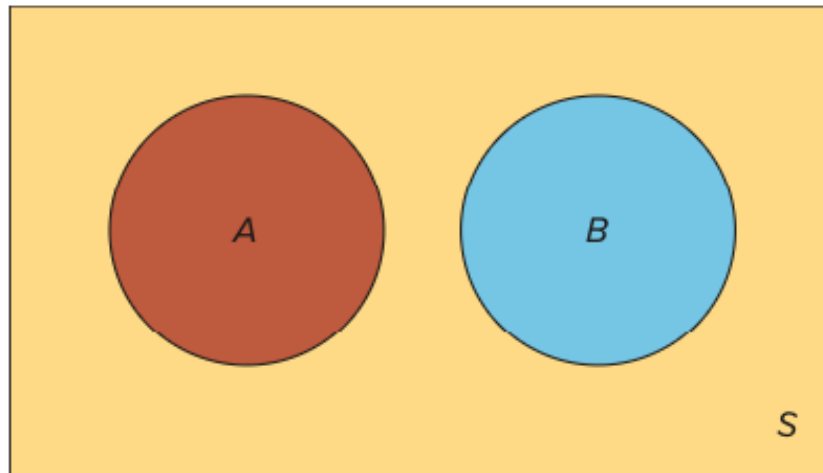
- Example: Anthony feels that he has a 75% chance of getting an A in Statistics and a 55% chance of getting an A in Managerial Economics. He also believes he has a 40% chance of getting an A in both classes.
 - a. What is the probability that he gets an A in at least one of these courses?
 - b. What is the probability that he does not get an A in either of these courses?

1: Probability Concepts and Probability Rules

- Let $P(A_S)$ correspond to the probability of getting an A in Statistics, $P(A_S) = 0.75$.
 - Let $P(A_M)$ correspond to the probability of getting an A in Managerial Economics, $P(A_M) = 0.55$.
 - Also, $P(A_S \cap A_M) = 0.40$.
- a.
$$\begin{aligned} P(A_S \cup A_M) &= P(A_S) + P(A_M) - P(A_S \cap A_M) \\ &= 0.75 + 0.55 - 0.40 \\ &= 0.90. \end{aligned}$$
- b. The probability that he does not receive an A in either of these two courses is the complement of the union; that is, $P((A_S \cup A_M)^c)$. So using the complement rule we have
- $$P((A_S \cup A_M)^c) = 1 - P(A_S \cup A_M) = 1 - 0.90 = 0.10.$$

1: Probability Concepts and Probability Rules

- For mutually exclusive events A and B , the probability of their intersection is zero $P(A \cap B) = 0$.



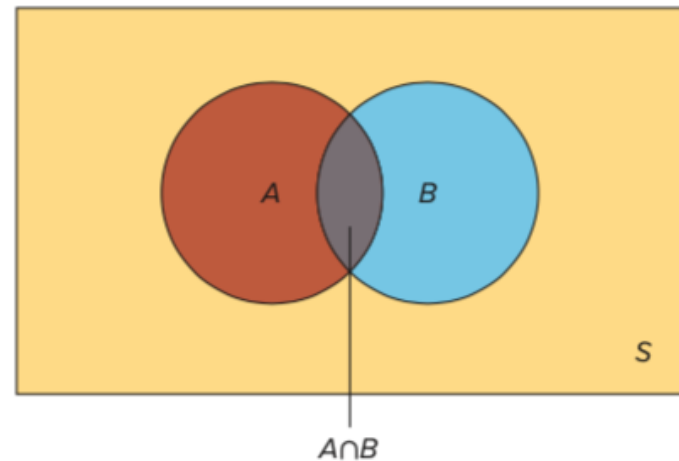
- There is no double counting
- $P(A \cup B) = P(A) + P(B)$

2: Conditional Probability Concepts

- Suppose the probability that a recent business college graduate finds a suitable job is 0.80.
 - Let A represents finding a job, $P(A) = 0.80$.
 - This is an unconditional probability.
- The probability of finding a suitable job is 0.90 if the recent business graduate has prior work experience.
 - Let B represent prior work experience, $P(A|B) = 0.90$.
 - The probability of an event is conditional on the occurrence of another event, so it is a conditional probability.

2: Conditional Probability Concepts

- Given two events A and B, each with a positive probability of occurring, the probability that A occurs given that B has occurred is derived as $P(A|B) = \frac{P(A \cap B)}{P(B)}$.
 - Because $P(A|B)$ is conditional on B (B has occurred), the sample space reduces to B.
- Similarly, $P(B|A) = \frac{P(A \cap B)}{P(A)}$.



2: Conditional Probability Concepts

Example: An economist predicts a 60% chance that country A will perform poorly and a 25% chance that country B will perform poorly. There is also a 16% chance that both countries will perform poorly.

- What is the probability that country A performs poorly given that country B performs poorly?

2: Conditional Probability Concepts

- Let A represent the event that country A performs poorly, $P(A) = 0.60$.
- Let B represent the event that country B performs poorly, $P(B) = 0.25$.
- The probability that country A performs poorly and country B performs poorly is $P(A \cap B) = 0.16$.
- The probability that country A performs poorly given country B performs poorly is $P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.16}{0.25} = 0.64$.
- The conditional probability is greater than the unconditional probability, so country B performing poorly exerts a positive influence on country A performing poorly.

2: Conditional and Joint Probability

- We can find the joint probability as the product of probabilities using the conditional probability formula; this is *the multiplication rule*.
- The joint probability of events A and B is derived as

$$P(A \cap B) = P(A|B)P(B).$$

- Example: A manager believes that 14% of consumers will respond positively to the firm's social media. Also, 24% of those who respond positively will become loyal customers.
- What is the probability that the next recipient of their social media campaign will react positively and become a loyal customer?

2: Conditional and Joint Probability

- Let R represent the event that a consumer who responds positively to a social media campaign, so $P(R) = 0.14$.
- Let L represent the event that a consumer is loyal, so $P(L|R) = 0.24$.
- The probability that a consumer responds positively to a social media campaign and is loyal is given by $P(R \cap L) = P(L|R)P(R) = 0.24 * 0.14 = 0.0336$.

2: Independent and Dependent Events

- Events are considered **dependent** if the occurrence of one is related to the probability of the occurrence of the other event.
- Two events are **independent** if the occurrence of one event does not affect the probability of the occurrence of the other event.
- Two events, A and B, are independent if
 - $P(A|B) = \frac{P(A)P(B)}{P(B)} = P(A)$
 - where $P(A \cap B) = P(A)P(B)$.

2: Independent and Dependent Events

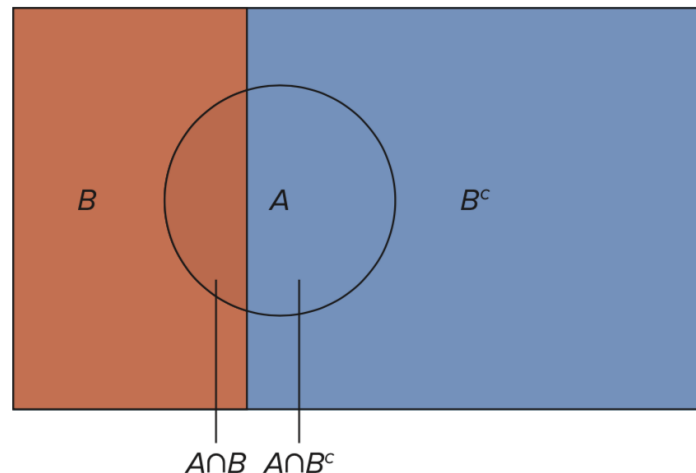
- **Example:** Suppose that for a given year there is a 2% chance that your desktop computer will crash and a 6% chance that your laptop computer will crash. Moreover, there is a 0.12% chance that both computers will crash. Is the reliability of the two computers independent of each other?

2: Independent and Dependent Events

- Let D represent the outcome that your desktop crashes, $P(D) = 0.02$.
- Let L represent the outcome that your laptop crashes, $P(L) = 0.06$.
- The joint probability is $P(D \cap L) = 0.0012$.
- Calculate $P(D|L) = \frac{P(D \cap L)}{P(L)} = \frac{0.0012}{0.06} = 0.02$.
- So, $P(D|L) = P(D)$. If your laptop crashes, it does not alter the probability that your desktop also crashes.
- The reliability of the two computers is independent.

2: The Total Probability Rule and Bayes' Theorem

- The total probability rule expresses the probability of an event, A , in terms of probabilities of the intersection of A with any mutually exclusive and exhaustive events.
- The total probability rule based on two events, B and B^c , is given by $P(A) = P(A \cap B) + P(A \cap B^c)$.



2: The Total Probability Rule and Bayes' Theorem

- **Bayes' theorem** is a procedure for updating probabilities based on new information; it uses the total probability rule.
- The original probability is an unconditional probability called a **prior probability**, in the sense that it reflects only what we know before the arrival of new information.
- On the basis of new information, we update the prior probability to arrive at a conditional probability called a posterior probability.
- The **posterior probability** $P(B|A)$ can be found using the information on the prior probability $P(B)$ along with conditional probabilities as

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A \cap B)}{P(A \cap B) + P(A \cap B^c)} = \frac{P(A|B)P(B)}{P(A|B)P(B) + P(A|B^c)P(B^c)}.$$

2: The Total Probability Rule and Bayes' Theorem

Example:

- In a lie-detector test, an individual is asked to answer a series of questions while connected to a polygraph (lie detector).
- This instrument measures and records several physiological responses of the individual on the basis that false answers will produce distinctive measurements.
- Assume that 99% of the individuals who go in for a polygraph test tell the truth.
- Assume the test is 95% reliable, e.g. 95% chance that the test will detect a lie if an individual actually lies.
- Let there also be a 0.5% chance that the test erroneously detects a lie even when the individual is telling the truth.

An individual has just taken a polygraph test and the test has detected a lie. What is the probability that the individual was actually telling the truth?

2: The Total Probability Rule and Bayes' Theorem

- Let D and T correspond to the events that the polygraph detects a lie and that an individual is telling the truth, $P(T) = 0.99$ and $P(T^c) = 0.01$.
- We formulate $P(D|T^c) = 0.95$ and $P(D|T) = 0.005$.
- We can use Bayes' theorem to find

$$P(T|D) = \frac{P(D|T)P(T)}{P(D|T)P(T) + P(D|T^c)P(T^c)} = \frac{0.005 * 0.99}{0.005 * 0.99 + 0.95 * 0.01} = 0.34256.$$
- Although we can use this formula directly, we can use the below table to help solve the problem systematically.

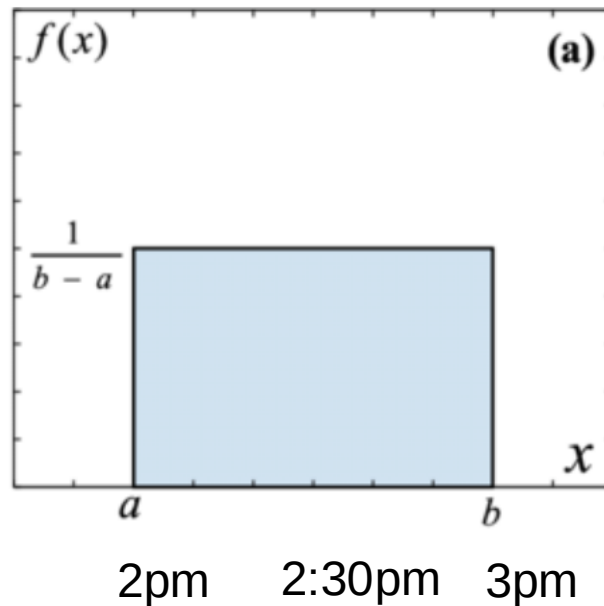
Prior Prob. (Given)		Conditional Prob. (Given)		Total Prob. For detect, P(D)		Posterior Prob.	
P(T)	0.99	P(D T)	0.005	P(D∩T)	0.005	P(T D)	0.343
P(Tc)	0.01	P(D Tc)	0.95	P(D∩Tc)	0.010	P(Tc D)	0.657
P(T)+P(Tc)	1	∩		P(D)	0.014	P(T D)+P(Tc D)	1.000

Random Variables and Probability Distributions

- A **random variable** is a function that assigns numerical values to the outcomes of an experiment.
 - A discrete random variable assumes a countable number of distinct values.
 - A continuous random variable is characterized by an uncountable values within an interval.
- Let look at a simple probability distribution for a continuous random variable called the **continuous uniform distribution**.
 - Uniform distribution $\leftarrow$ the underlying random variable has an **equally likely chance of assuming a value** within a specified range.

3: The probability distribution

- Example: Suppose you are informed that your new refrigerator will be delivered between 2:00 pm and 3:00 pm.
 - Let X denote the delivery time, this is bounded below by 2:00 pm and above by 3:00 pm for a total range of 60 minutes.
 - The probability of a delivery between 2:00 pm and 2:30 is $30/60 = 0.50$.

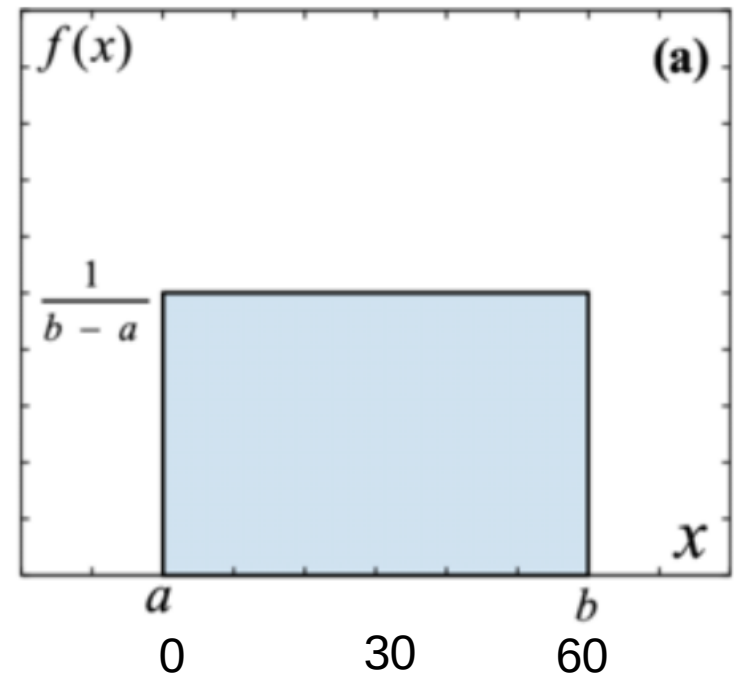


A math Approach for Continuous Distribution

- A continuous random variable is completely described by its **probability density function**, denoted $f(x)$.
 - The area under $f(x)$ over all values of X must equal one.
 - The probability the variable assumes a value with an interval, $P(a \leq X \leq b)$ is defined as the area under $f(x)$ between points a and b .

$$\int_a^b f(x) \cdot dx$$

- $\text{Prob}(X \leq 30) = \int_0^{30} \frac{1}{60} * dx = \frac{1}{60} * (30 - 0)$



3: The Normal Distribution

- The **normal distribution** is one of the most widely used continuous distributions.
 - Closely approximates the probability distribution for a wide range of random variables
 - **Bell-shaped and symmetric** around its mean
 - Gaussian distribution
- A graph depicting the normal **probability density function** $f(x)$ is often referred to as the normal curve or bell curve.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

- Generally use the **cumulative distribution function** $P(X \leq x)$ to compute probabilities: the area under the normal curve up to the value x .
- Use tables or software to find probabilities.

3: The Normal Distribution

- The **standard normal distribution** is a special case of the normal distribution, denoted by a letter Z .
 - The mean is $E(Z) = 0$.
 - The standard deviation is $SD(Z) = 1$.
- A standard normal table, also referred to as a z table, provides areas (probabilities) under the z curve for $P(Z \leq z)$.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	↑
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	↑
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
0.4	←	←	←	←	←	←	←	0.6808

- The first column is for the ones and tenths place of a z value.
- The other columns are for the hundredths place of a z value.
- Where these intersect is $P(Z \leq z)$.
- There is one page for negative values of z and another page for positive values of z .

3: The Normal Distribution

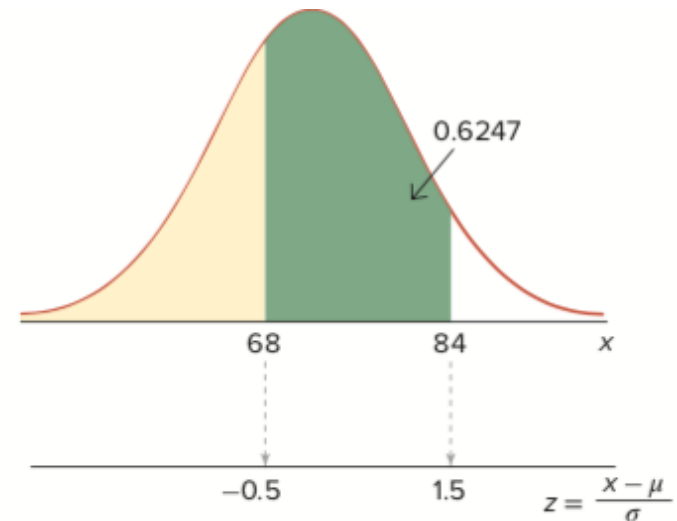
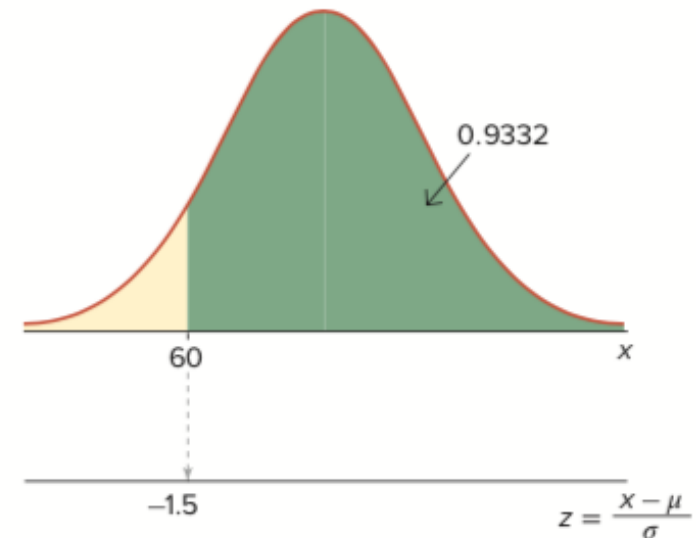
- For a general normal distribution, $P(X \leq x)$ requires integral calculus, or certain calculators or statistical packages.
- However, we can transform any normally distributed random variable into the standard normal random variable and use the tables.
- Let X have a normal distribution with mean μ and standard deviation σ . X can be transformed into Z using $Z = \frac{X - \mu}{\sigma}$.
- Example: Scores on a management aptitude exam are normally distributed with a mean of 72 and a standard deviation of 8.
 - a. What is the probability that a randomly selected manager will score above 60?
 - b. What is the probability that a randomly selected manager will score between 68 and 84?

3: The Normal Distribution

- Let X represent scores with $\mu = 72$ and $\sigma = 8$.

a.
$$P(X > 60) = P\left(Z > \frac{60-72}{8}\right) = P(Z > -1.5) = 1 - P(Z \leq -1.5) = 1 - 0.0668 = 0.9332$$

b.
$$P(68 \leq X \leq 84) = P\left(\frac{68-72}{8} \leq Z \leq \frac{84-72}{8}\right) = P(-0.5 \leq Z \leq 1.5) = P(Z \leq 1.5) - P(Z < -0.5) = 0.9332 - 0.3085 = 0.6247.$$



Conclusions

- Concepts and rules of probability are explained, e.g. events, experiments and Complement.
- The conditional probability and Bayes' theorem are explained.
- Describe the normal distribution and calculate its associated probabilities,
 - Incl. the use of z-table.

Reference

- Z-table <http://www.z-table.com>